

RADIOACTIVE CLOCKS AND COSMIC-RAY TRANSPORT IN THE GALAXY

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Abstract. There are a number of radioactive ‘clocks’ in the cosmic radiation that can be used to measure the time scales for cosmic ray processes in the Galaxy. With high-resolution isotope measurements available from ACE it is now possible to read these clocks with greatly improved accuracy and address key questions about the origin and lifetime of cosmic rays. This paper discusses the status of three such investigations.

1. Introduction

It is now generally agreed that the only sources capable of providing the 3×10^{40} ergs per second necessary to sustain cosmic rays in the Galaxy are supernova (SN) explosions. It is also generally accepted that the mechanism by which the bulk of cosmic rays are energized is shock acceleration – the same mechanism that apparently accelerates most of the energetic particles that originate in the heliosphere. Beyond this, there remain several outstanding questions as to the origin of cosmic rays. They include: What is the origin of the material that gets accelerated? Do cosmic rays originate from fresh supernova ejecta, ‘old’ interstellar material, or a combination of these? Are cosmic rays accelerated by a ‘one-shot’ mechanism (i.e., by a single SN shock), or by distributed acceleration continuing over their lifetime (i.e., a series of SN shocks)? What volume of the Galaxy do cosmic rays fill? Does it include a galactic halo as well as the disk?

These questions are all related to the time scales of cosmic-ray processes in the Galaxy, including the elapsed time between the nucleosynthesis of cosmic-ray material and cosmic-ray acceleration, the time scale for the acceleration process(es), and the time scale over which cosmic rays are stored in the Galaxy. Radioactive isotopes in cosmic rays provide a means of measuring these time scales.

Cosmic rays can be divided into two general classes. *Primary cosmic rays* are those that survive after being accelerated from the original pool of source material. *Secondary cosmic rays* are produced by nuclear interactions of (mostly) primary



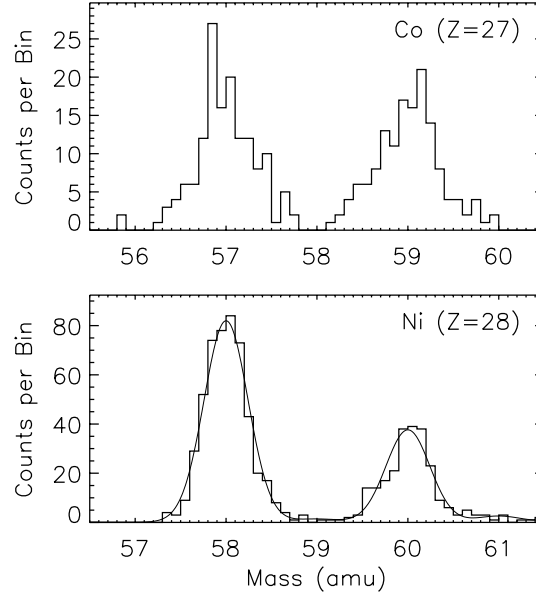


Figure 1. Mass histograms of Co and Ni with ~ 100 to 500 MeV amu^{-1} measured by the CRIS experiment on ACE (Wiedenbeck *et al.*, 1999). Note the relative absence of ^{59}Ni and the presence of ^{59}Co .

cosmic rays with the interstellar medium (ISM). Secondary-to-primary ratios such as $^3\text{He}/^4\text{He}$, B/C and $(21 \leq Z \leq 23)/\text{Fe}$ are used to measure the amount of material traversed by cosmic rays in the Galaxy (commonly called the pathlength). Most primaries are stable; among the few radioactive species that may survive are the actinides Th, U, and possibly Np and Cm. Three primary isotopes in the Fe region that decay mainly by electron capture may also survive, depending on the time-delay between nucleosynthesis and cosmic ray acceleration. In addition, there are radioactive secondaries that can be used to measure the cosmic-ray lifetime (^{10}Be , ^{26}Al , ^{36}Cl , and ^{54}Mn), and electron-capture nuclei that can be used as densitometers, and/or as tests of whether cosmic rays are continuously reaccelerated during their lifetime.

In 1997 the Advanced Composition Explorer (ACE) was launched carrying a suite of instruments designed to measure the composition of energetic particles with much greater precision than was possible previously (Stone *et al.*, 1998a). Included was a Cosmic Ray Isotope Spectrometer (CRIS) that has achieved excellent mass resolution ($\sigma_m = 0.1$ to 0.25 amu for $2 \leq Z \leq 30$) with the collecting power ($\sim 250 \text{ cm}^2\text{sr}$) needed to resolve even rare species (Stone *et al.*, 1998b). In this paper we discuss three examples of cosmic rays clocks for which considerable progress has been achieved with ACE data, and consider the implications of these new measurements for our understanding of cosmic ray acceleration and transport in the Galaxy.

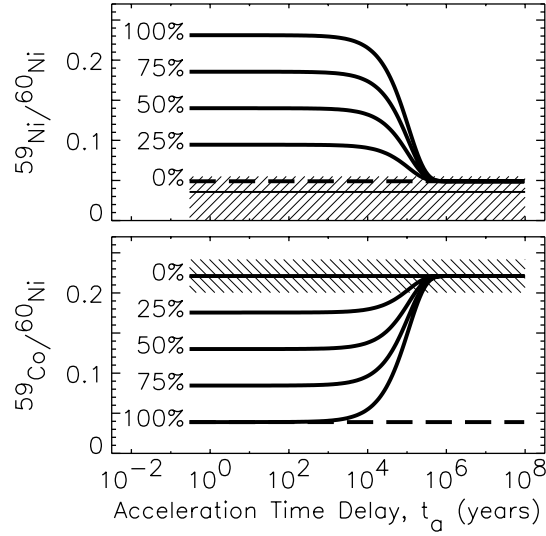


Figure 2. Allowed combinations of acceleration delay times and fractions of mass-59 material synthesized as ^{59}Ni , as derived from ^{59}Ni and ^{59}Co measurements on ACE (Wiedenbeck *et al.*, 2000). The cross-hatched region shows 98% confidence limits based on ACE data alone; the diagonally-hatched region also takes into account cross section uncertainties.

2. Acceleration-Time Delay Clocks

Some time ago, Soutoul *et al.* (1978) identified three electron-capture nuclei in the Fe-group (^{59}Ni , ^{57}Co , and ^{56}Ni), all products of explosive nucleosynthesis, that essentially decay only by electron-capture (EC). Since cosmic rays with >500 MeV/nucleon spend most of their time as fully-stripped nuclei, a significant fraction of these nuclei would be expected to survive if cosmic rays are accelerated to high energies soon after nucleosynthesis. Soutoul *et al.* (1978) suggested that the abundances of ^{56}Ni , ^{57}Co , and ^{59}Ni , with electron capture half-lives ranging from ~ 6 days to $\sim 10^5$ yr., could provide a measure of the time delay between nucleosynthesis and cosmic-ray acceleration. It is evident that ^{56}Ni has decayed in cosmic rays because the dominant Fe-peak isotope is ^{56}Fe , as in the solar system. The more interesting cases are ^{57}Co ($T_{1/2} = 0.74$ y) and ^{59}Ni ($T_{1/2} = 7.6 \times 10^4$ y). The Ni and Co isotope measurements in Figure 1 (Wiedenbeck *et al.*, 1999) show that there is very little ^{59}Co in cosmic rays, suggesting that a reasonably long time delay has occurred.

However, we must also consider the possibility that a significant fraction of the mass 59 nuclei were synthesized as ^{59}Co instead of ^{59}Ni . Figure 2, from Wiedenbeck *et al.* (2000) indicates that the time delay must have been $\geq 3 \times 10^4$ years as long as at least 20% of the mass 59 material originated as ^{59}Ni , and $\geq 10^5$ years if at least 40% of the mass-59 nuclei originated as ^{59}Ni . Woosley and Weaver (1995) have calculated the yields of Fe-group isotopes from Type II supernova. For

stellar masses ranging from 11 to 25 solar masses, the fraction of mass 59 nuclei originating as ^{59}Ni ranges from 27% to 87%. For a Salpeter initial mass function it is found that $\sim 68\%$ of the mass 59 material originates as ^{59}Ni (Wiedenbeck *et al.*, 1999). It therefore appears that only time delays $>10^5$ years are allowed unless cosmic rays come from a very narrow range of stellar masses (in which case isotope anomalies would be expected in other Fe-group elements).

A time delay of $\geq 10^5$ years between nucleosynthesis and acceleration is inconsistent with models in which supernovae accelerate their own ejecta. It is consistent with models in which cosmic rays are accelerated from the interstellar medium (e.g., Olive and Schramm, 1982), or from a low-energy seed population pre-accelerated by stellar flares (Meyer, 1985). One would also expect a long time delay in models in which the refractory elements in cosmic rays originate from interstellar dust grains (Meyer *et al.*, 1997). Higdon *et al.* (1998) have suggested that most cosmic rays originate in super-bubbles in which a series of SN explosions extend over periods of $\sim 10^5$ years or more. In this case, for ^{59}Ni to decay before acceleration, it is necessary that the ejecta from SN early in the life of the super-bubble remains relatively un-mixed with the surrounding ISM for at least $\sim 10^5$ years before being accelerated to high energies by shock waves from subsequent SN.

It is also possible that cosmic ray ^{59}Ni nuclei could pick up electrons and decay in-flight if they manage to retain these electrons for a time comparable to the EC-decay lifetime of 10^5 years. The average energy of the CRIS measurements is $\sim 300 \text{ MeV nucleon}^{-1}$, corresponding to an average of ~ 400 to $500 \text{ MeV nucleon}^{-1}$ outside the heliosphere. Wiedenbeck *et al.* (2000) show that at energies of $450 \text{ MeV nucleon}^{-1}$ or more the electron attachment mean free path (mfp) is several g/cm^2 , while the electron stripping mfp is $< 10^{-2} \text{ g cm}^{-2}$. At these energies one does not expect appreciable ^{59}Ni decay for the mean density ~ 0.3 to 0.4 cm^{-3} indicated by the ^{10}Be and other beta-decay clock measurements (see below). However, if the observed cosmic rays have spent an appreciable time at lower energies (i.e., $< 200 \text{ MeV nucleon}^{-1}$, where the electron-capture mfp is $< 1 \text{ g cm}^{-2}$) before getting accelerated to higher energies, it is possible that a significant fraction of the ^{59}Ni could have decayed in-flight. The question of whether cosmic rays are accelerated in a step-wise fashion (also called re-acceleration) is discussed below.

3. Electron-Capture Nuclei as Probes of Cosmic Ray Transport Processes

Over the $\sim 10^4$ year lifetime of a single SN blast wave it is theoretically possible to accelerate cosmic ray nuclei to $\sim 10^{14}Z \text{ MeV}$, where Z is the atomic number (see, e.g., Gaisser, 1990). However, it is also possible that most cosmic rays gain their energy from a series of SN shock waves. One possible way to address this question is with measurements of electron-capture nuclei that would be expected to decay if the observed cosmic rays have spent an appreciable fraction of their

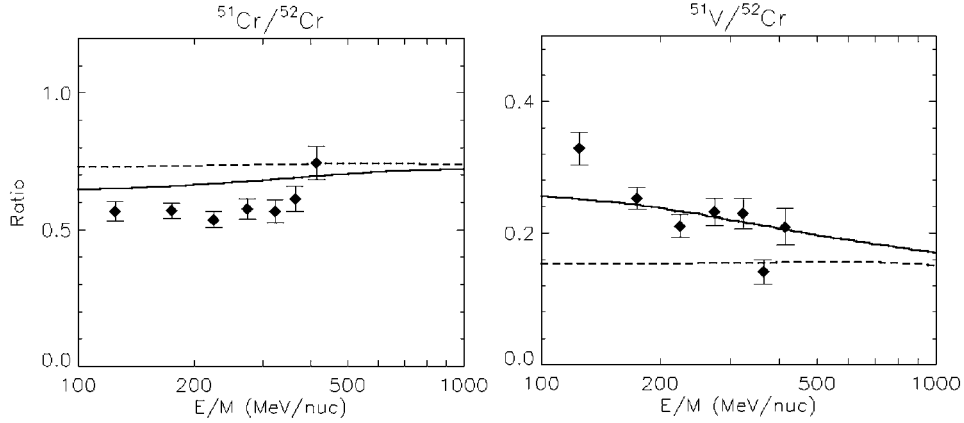


Figure 3. Energy dependence of two isotope ratios that demonstrate energy-dependent electron-capture decay of ^{51}Cr to ^{51}V (Niebur *et al.*, 2000). The curves show the expected ratios with (solid) and without decay (dashed). The sum of $(^{51}\text{Cr}+^{51}\text{V})/^{52}\text{Cr}$ is found to be energy independent.

lifetime at lower energies where electron attachment is much more likely. Nuclides that can probe these processes include ^7Be , ^{37}Ar , ^{44}Ti , ^{49}V , ^{51}Cr , ^{55}Fe , and ^{57}Co , with laboratory lifetimes ranging from 28 days to 67 years.

Connell and Simpson (1999), Connell (2001) and Soutoul *et al.* (1998) have concluded that a significant fraction of ^{49}V and ^{51}Cr have decayed, based on comparisons of *Ulysses* and *Voyager* measurements of the $^{49}\text{V}/^{51}\text{V}$ ratio with a leaky-box model (LBM) of cosmic-ray propagation in the Galaxy. They conclude that some re-acceleration has occurred.

With the large collecting power of CRIS it is possible to examine the abundances of EC-nuclei in several energy intervals. In the results of Niebur *et al.* (2000), shown in Figure 3, there appears to be a clear deficit of the EC-nuclide ^{51}Cr at lower energies, and a corresponding overabundance of the daughter ^{51}V . When the sum of ^{51}Cr and ^{51}V are compared with stable secondary species there is no observed energy dependence. Similar results are found for the parent-daughter combination ^{49}V and ^{49}Ti (Niebur *et al.*, 2000). These results support the conclusion that the observed variation of these ratios with energy is due to the energy-dependence of electron-capture and loss processes rather than some unexpected energy dependence of Fe-fragmentation cross sections.

In contrast to the findings of Connell and Simpson (1999) and Soutoul *et al.* (1998), the energy dependence of the ^{51}Cr and ^{51}V abundances in Figure 3 (and also those of ^{49}V and ^{49}Ti ; see Niebur *et al.* (2000), appear to be reasonably well accounted for without invoking reacceleration. In addition, Webber (2000a) concludes that the $^{51}\text{V}/^{49}\text{V}$ ratio measured by ACE is consistent with a LBM without reacceleration. Webber (2000b) has also compared ACE and *Ulysses* data with a Monte Carlo diffusion model and finds that the $^{51}\text{V}/^{51}\text{Cr}$ data favor reacceleration, while the $^{49}\text{Ti}/^{49}\text{V}$ data do not.

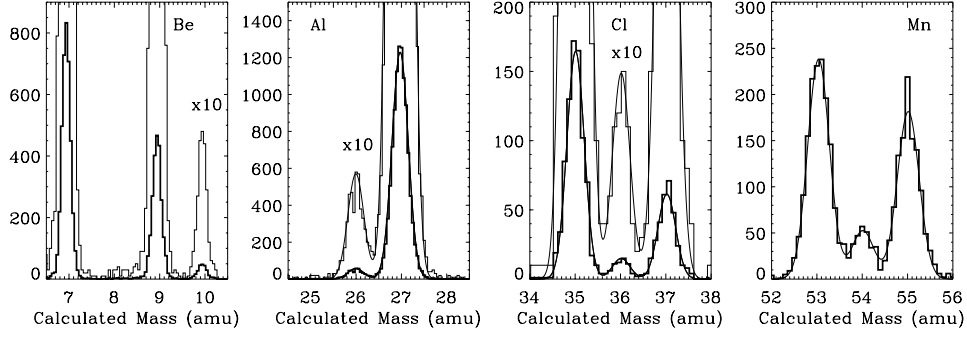


Figure 4. Mass histograms of Be, Al, Cl, and Mn isotopes measured by the CRIS experiment on ACE (see Yanasak *et al.*, 2001).

Observations of EC isotopes in cosmic rays are still in their infancy, as are models that interpret them. Their abundances depend on the energy dependence of the electron-capture and loss cross sections and the fragmentation cross sections, on the history of cosmic ray acceleration and possible re-acceleration, on inhomogeneities in the interstellar matter traversed by cosmic rays (Raisbeck *et al.*, 1975; Letaw *et al.*, 1984), and on energy changes that occur as cosmic rays enter the heliosphere. It is therefore premature to draw firm conclusions from these new observations without examining thoroughly the effects of uncertainties in modeling these processes. However, it is already clear that these data will be the subject of lively debate over the next few years.

4. Beta-Decay Clocks and Cosmic-Ray Transport in the Galaxy

The lifetime of cosmic rays in the Galaxy is important because it determines the power required to sustain the energy density of cosmic rays in the Galaxy (e.g., Gaisser, 1990). This lifetime can be derived from the measurements of radioactive isotopes such as ^{10}Be (half-life of 1.5 My). The first reliable observations of ^{10}Be in cosmic rays occurred during the 1970s, when it was discovered that most ^{10}Be produced in cosmic-ray interactions has decayed, leading to an estimated confinement time of $\tau_{\text{esc}} \approx 10$ to 30 My for cosmic-ray storage in the Galaxy (see review in Simpson and Garcia-Munoz, 1988). Taking into account that cosmic rays with several hundred MeV amu^{-1} have traversed $\sim 5 \text{ g cm}^{-2}$ of interstellar material, this lifetime suggests that the mean density of the medium that cosmic rays inhabit is ~ 0.2 to $\sim 0.4 \text{ H atoms cm}^{-3}$, substantially less than the mean density commonly ascribed to the galactic disk ($\sim 1 \text{ H atom cm}^{-3}$). This has led to suggestions that cosmic rays must spend appreciable time in low-density regions such as the galactic halo.

With the advent of the *Ulysses* and *ACE* missions it has become possible to accurately measure and compare abundances of four separate beta-decay clocks

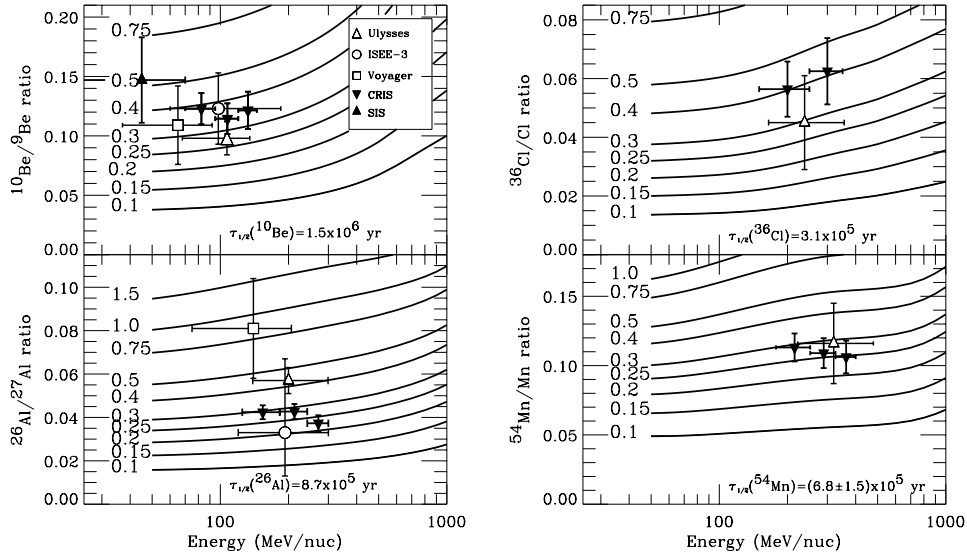


Figure 5. Isotope ratios involving four cosmic-ray clocks are compared to predictions of the leaky box model, parameterized by the density of the confinement region in H atoms per cm^3 (Yanasak *et al.*, 2001). Measurements are from *ACE* (Yanasak *et al.*, 2001); *Ulysses* (Connell, 1998; Connell *et al.*, 1998; Simpson and Connell, 1998; Du Vernois, 1997); *Voyager* (Lukasiak *et al.*, 1994, 1997a, and 197b), and *ISEE-3* (Wiedenbeck and Greiner, 1980).

TABLE I
Radioactive clock parameters^a

Isotope	Half-life (My)	Density (H atoms per cubic cm)	Surviving fraction	Confinement time (My)
^{10}Be	1.51	$0.358 \pm .032 \pm .058$	$0.252 \pm .018 \pm .039$	$14.5 \pm 1.3 \pm 2.2$
^{26}Al	0.873	$0.276 \pm .018 \pm .084$	$0.248 \pm .012 \pm .056$	$20.4 \pm 1.3 \pm 5.7$
^{36}Cl	0.307	$0.425 \pm .067 \pm .077$	$0.155 \pm .021 \pm .024$	$13.8 \pm 2.2 \pm 2.6$
^{54}Mn	$0.63 \pm .17$	$0.267 \pm .027 \pm .084, -.060$	$0.284 \pm .019 \pm .022$	$22.4 \pm 2.3 \pm 6.1$

^aUncertainties are shown as (a) statistical only; and (b) systematic uncertainties added in quadrature. These results are from Yanasak *et al.* (2001).

(^{10}Be , ^{26}Al , ^{36}Cl , and ^{54}Mn) with substantially different half-lives. Measurements of these clock isotopes from *ACE* are shown in Figure 4. A summary of *ACE*, *Ulysses*, *Voyager*, and *ISEE-3* measurements is shown in Figure 5. Also shown are calculations for a steady state LBM parameterized by the mean density of matter in the cosmic ray storage region (Yanasak *et al.*, 2000). Note that there is generally good agreement, and that the *ACE* results for all four clocks are consistent with an average density of $0.36 \text{ H atoms cm}^{-3}$. These calculations employ an energy-dependent mean pathlength of a form suggested by Soutoul and Ptuskin (1999),

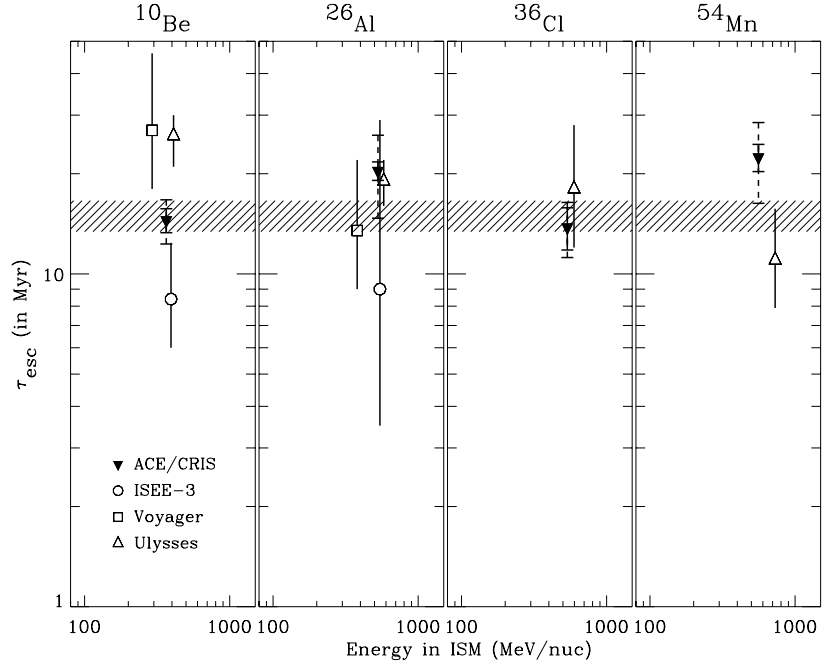


Figure 6. Cosmic ray escape lifetimes measured by *ACE*, *Ulysses*, *Voyager*, and *ISEE-3*. For references see the caption to Figure 5.

$$\Lambda_{\text{esc}}(\text{g cm}^{-2}) = 26.7\beta/[(\beta R/1.0 \text{ GV})^{0.6} + (\beta R/1.4 \text{ GV})^{-1.4}], \quad (1)$$

that provides a good fit to the B/C and (Sc + Ti + V)/Fe ratios (Davis *et al.*, 2000; Yanasak *et al.*, 2001) using cross-sections of Silberberg *et al.* (1998), an interstellar He/H ratio of 0.1, and an ionized hydrogen fraction of 10.5%. In the LBM the mean lifetime against escape from the Galaxy is $\tau_{\text{esc}} = \Lambda_{\text{esc}}/(\beta c \rho)$, where ρ is the interstellar density in g cm^{-3} and the velocity, β , in units of the speed of light (c) is evaluated at an energy that accounts for energy loss during solar modulation. Table I summarizes these and other derived quantities. For a discussion of systematic uncertainties, see Yanasak *et al.* (2001).

Figure 6 compares τ_{esc} values from *ACE*, *Ulysses*, *Voyager*, and *ISEE-3*; all based on models with somewhat different cross sections and parameters. Although there is general agreement, the ^{10}Be results differ considerably, and the mean value of $\tau_{\text{esc}} = 14.5 \pm 1.5 \text{ My}$ from *ACE* data is somewhat less than that derived from *Ulysses* data (Connell, 2001). The difference between the *ACE* and *Ulysses* results for ^{54}Mn is result of the lower abundance of Fe secondaries relative to Fe reported by *Ulysses* (see Lijowski *et al.*, 1999). This ratio affects the derived value of Λ_{esc} , and therefore also of τ_{esc} . Note that the *ACE* points show separately statistical uncertainties and also uncertainties due to cross sections, solar modulation, and the ionized fraction of the ISM (Yanasak *et al.*, 2001). Other studies have not usually included these systematic uncertainties.

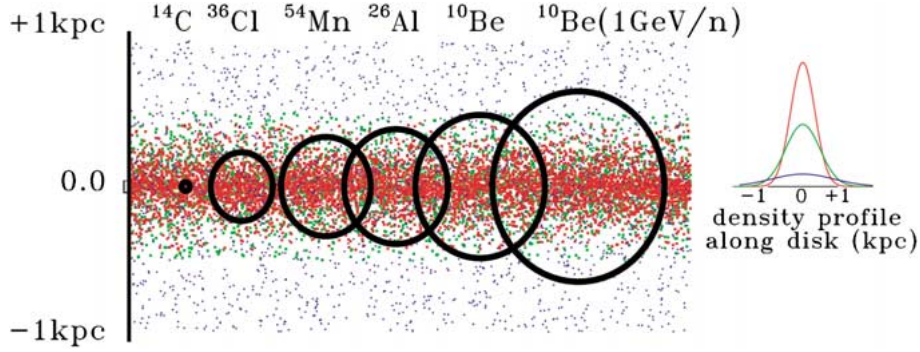


Figure 7. Schematic illustration of the size of the region sampled by several radioactive clocks, compared to the scale height of the interstellar gas assumed by Ptuskin and Soutoul (1998). The first five circles are representative of the typical energies measured by *ACE*, and assume that the mean distance a radioactive isotope can arrive from is given by $R = (\gamma DT_d)^{1/2}$, where D is the diffusion coefficient (assumed to be $2 \times 10^{28} \text{ cm}^2 \text{ s}^{-1}$) and T_d is the mean life for decay. The sixth circle is for $1 \text{ GeV amu}^{-1} {}^{10}\text{Be}$.

One goal of this study is to determine the characteristics of regions where cosmic rays reside. In the LBM the cosmic-ray intensity and ISM density are uniform in space and time, and the surviving fractions of various species are sensitive only to the mean density of the storage region. There are now more realistic models, including diffusion models in which cosmic rays also have access to a galactic halo or other low-density regions (e.g., Webber *et al.*, 1992; Strong and Moskalenko, 1998; Ptuskin and Soutoul, 1998; Streitmatter and Stephens, 2000). In such models the distance (R) from which radioactive species can arrive is limited by its half-life, such that $R \approx (\gamma DT_d)^{1/2}$, where D is the diffusion coefficient, T_d is the mean life for radioactive decay at rest, and $\gamma = 1/(1 - \beta^2)^{1/2}$. In this picture the abundance of a given radioactive clock is sensitive mainly to conditions within a sphere around the Sun whose radius scales as R .

Figure 7 illustrates schematically the size of the sphere sampled by five radioactive superimposed on a representation of the density profile of the galactic disk. Note that the ${}^{10}\text{Be}$ measurements shown here reflect conditions out to $\sim 400 \text{ pc}$, well beyond the disk, while the shorter-lived species probe shorter distances. Although there are as yet no reliable measurements of ${}^{14}\text{C}$ in cosmic rays, any observed ${}^{14}\text{C}$ can originate only in very nearby regions. The only approach currently available to extend this study to greater distances is to extend ${}^{10}\text{Be}$ measurements to energies above $\sim 1 \text{ GeV amu}^{-1}$ where relativistic time-dilation effects become important.

If cosmic-ray clocks are indeed sampling a density distribution such as that in Figure 7 we can expect the average density determined from measurements of longer-lived species such as ${}^{10}\text{Be}$ to be less than that for shorter-lived species such as ${}^{36}\text{Cl}$, and the value of τ_{esc} to increase with increasing half-life. From the data in Figure 8 it appears that ${}^{36}\text{Cl}$ fits this picture relative to the other clocks, but

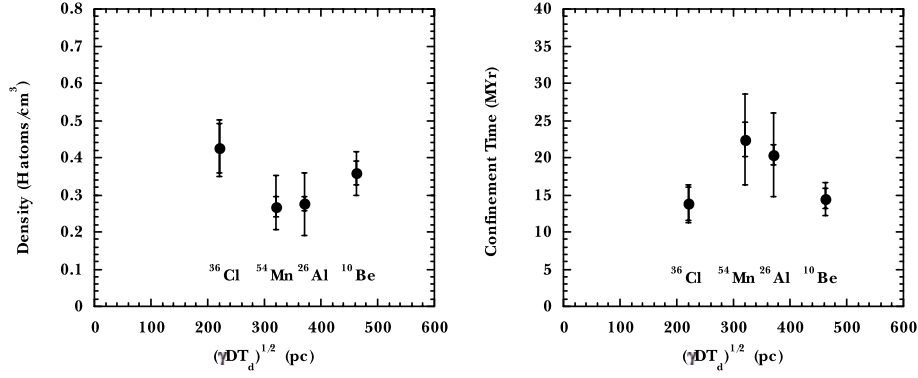


Figure 8. (a) Plot of interstellar densities (in H atoms cm^{-3}) derived from four clocks vs $R = (\gamma DT_d)^{1/2}$ (see text); (b) Confinement time vs. R . The smaller uncertainties are statistical only; the larger uncertainties include systematic uncertainties (see Table I).

the uncertainties in these derived quantities are currently too large to draw firm conclusions. Yanasak *et al.* (2001) has shown that uncertainties in the density and τ_{esc} are dominated by cross section uncertainties rather than by uncertainties in the cosmic-ray isotope ratios. For ⁵⁴Mn the uncertainty in its β -minus half-life dominates. As a result, it may be possible to learn more from these measurements if cross section uncertainties can be reduced and the ⁵⁴Mn β -minus half-life better determined. However, it should also be remembered that cosmic rays need not venture into the halo to find low-density regions (see, e.g., Diplas and Savage, 1993). It is possible that the low-density region apparently sampled by the cosmic rays we observe is not our galactic halo, but corresponds instead to the local bubble (see, e.g., Streitmatter and Stephens, 2001).

Ptuskin and Soutoul (1998) have calculated the expected surviving fraction of ¹⁰Be, ²⁶Al, and ³⁶Cl for four diffusion models that attempt to provide a more realistic picture of cosmic-ray transport than the LBM. The predictions are shown in Figure 9. The first model (a) has an infinitely thin disk. The second (b) includes diffusion in a disk with an exponential density distribution with scale height of 100 pc. In the third (c), the ISM is composed of three components with exponential scale heights of 130, 200, and 1000 pc (see Figure 6). The fourth (d) assumes the same gas distribution, but also places the solar system on the edge of a cavity of several hundred pc radius. Figure 9 also indicates the range of diffusion coefficients consistent with ACE measurements of these clocks. Note that in the more sophisticated of these models (c, d) all three clocks are consistent with $D \approx 2 \times 10^{28} \text{ cm}^2 \text{ s}^{-1}$.

Several authors have shown that discrimination between cosmic ray transport models is possible by extending measurements of ¹⁰Be to higher energies, where relativistic time dilation effects allow ¹⁰Be to arrive from greater and greater distances. Ptuskin (2000) has compared the surviving fraction of ¹⁰Be in the LBM, a static diffusion model, a ‘diffusive halo’ model with a galactic wind, and a

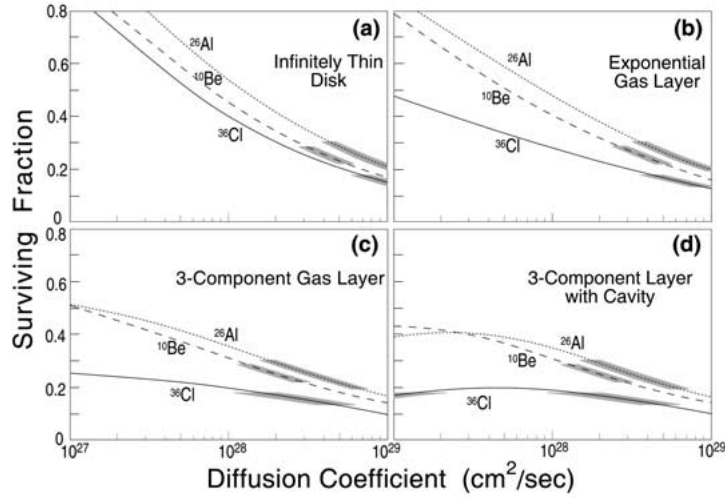


Figure 9. Expected surviving fractions of ^{10}Be , ^{26}Al , and ^{36}Cl at $400 \text{ MeV nucl}^{-1}$ for four diffusion models (Ptuskin and Soutoul, 1998), plotted versus the assumed diffusion coefficient. Values consistent with the *ACE* measurements are indicated.

model that includes diffusion and reacceleration. These diffusion models all predict that relatively less ^{10}Be will survive at higher energies than in the LBM. Webber (2000b) and Simon and Molnar (1999) find similar results. Available ^{10}Be measurements above $\sim 300 \text{ MeV amu}^{-1}$ do suggest a greater $^{10}\text{Be}/^9\text{Be}$ ratio than expected from the LBM (see Ahlen *et al.*, 2000; Buffington *et al.*, 1978, and references therein), although these studies suffer from limited statistical accuracy and/or mass resolution. Preliminary data from the ISOMAX magnetic spectrometer indicate that ^{10}Be has been resolved to $> 1 \text{ GeV amu}^{-1}$, but final results are not yet available.

Webber and Soutoul (1998) have attempted to estimate the surviving fraction of several radioactive clocks in the 1 to 20 GeV amu^{-1} region using abundance ratios such as Be/B , Al/Si , and Cl/S measured by HEAO-3 (Engelmann *et al.*, 1990). Taken at face value, their results suggest that the surviving fraction of ^{10}Be and other clock nuclei beyond 2 GeV amu^{-1} is less than expected in the LBM. However this approach is very sensitive to uncertainties in the fragmentation cross sections in an energy range where there are few measurements, and cross section differences of $\sim 20\%$ (as seen, e.g., in Yanasak *et al.*, 2001) would lead to significantly different conclusions. With direct measurements of relevant cross sections at 1 to 10 GeV amu^{-1} , this approach might prove fruitful.

5. Summary

This paper has summarized the first measurements of cosmic ray clocks from the CRIS instrument on ACE and discussed the implications for some key questions

in cosmic ray astrophysics. The decay of ^{59}Ni in cosmic rays implies that supernovae do not accelerate their own ejecta; rather, there must be a time delay $\geq 10^5$ years between the time of nucleosynthesis and acceleration. New measurements of ^{49}V and ^{51}Cr provide evidence for energy-dependent electron-capture decay, which may provide a new tool for studying energy-changing processes in the Galaxy and the heliosphere. Finally, *ACE* has now provided well-resolved measurements of four radioactive clocks (^{10}Be , ^{26}Al , ^{36}Cl , and ^{54}Mn) with greatly improved statistical accuracy. The observed abundances can be fit with a steady state leaky-box model characterized by an escape lifetime of 14.5 ± 1.5 My and a mean density of 0.36 H atoms cm^{-3} . However, these results are also consistent with a diffusive halo model characterized by a diffusion coefficient of $\sim 2 \times 10^{28}$ $\text{cm}^2 \text{s}^{-1}$. Although it is perhaps still too early to conclude that the clock has run out on the leaky box model, the focus of most theoretical efforts seeking to interpret the new, high precision observations is now directed towards more sophisticated models of cosmic ray transport.

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